

Digital Output MAPS Template (Instrumentation)

The Digital Output MAPS template is designed to control a digital output.

All MAPS templates provide the following:

- an associated PLC function block
- a set of SCADA tags (Adroit agents) linked to the provided signals
- an associated SCADA graphic with inbuilt faceplates

This MAPS template is available in three models – advanced, standard and basic, which provide respectively lower levels of control over the digital output.

This document describes the Advanced Digital Output MAPS Template and also defines the level of functionality provided by the standard and basic models. So that by reading this document you can decide which model best suits your needs. You should consider the following factors in your choice:

- the importance of the digital being controlled and
- the budget of the customer

Since the more complex the model, the greater the required PLC memory usage; number of PLC I/O and SCADA size (scan points) - see the **Resources** table below.

Advanced Digital Output Functional Philosophy

The digital output is enabled once an enable request is received from the PLC auto control program or an operator enables it from the faceplate.

Field Outputs

- Digital Output

Simulation mode - will disable the physical digital output and simulate the digital output in the PLC program. Simulation mode is used to test the process logic without enabling the physical output itself. It is very useful in testing, commissioning and maintenance.

Typical Faceplate Graphics for the Advanced Digital Output MAPS Template

Digital Output	
	
	Control / Tag name
	Manual mode indication
	Simulation mode indication
	Digital output status indication

Signal Description		Agent Type	Advanced	Standard	Basic
SCADA Control Words:					
SSCL 0	Control Word	Marshal	x	x	x
SSCL 1	Delay On Setpoint Time	Analog	x	x	
SSCL 2	Delay Off Setpoint Time	Analog	x		

Signal Description		Agent Type	Advanced	Standard	Basic
SCADA Status Words:					
SSSL 0	Status Word	Marshal	x	x	x

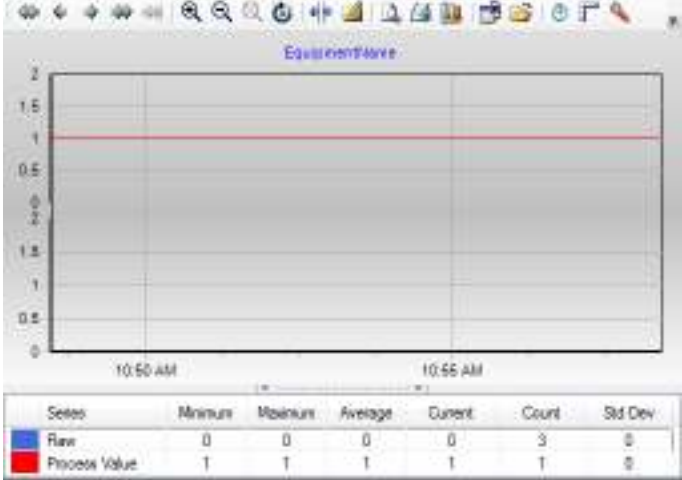
Digital Output:					
DO 0	Digital Output		x	x	x

SCADA Control Word (Adroit Marshal Agent)		Event Logged	Advanced	Standard	Basic
Bit 0	Output inverted	x	x	x	x
Bit 1	Auto/Manual Mode	x	x	x	x
Bit 2	Spare				
Bit 3	Simulation Enabled	x	x	x	x
Bit 4	Simulation Value	x	x	x	x
Bit 5	Alarm Enabled	x	x	x	x
Bit 6	Alarm on bit off	x	x	x	x
Bit 7	Alarm on bit on	x	x	x	x
Bit 8	Spare				
Bit 9	Spare				
Bit 10	Spare				
Bit 11	Spare				
Bit 12	Spare				
Bit 13	Spare				
Bit 14	Spare				
Bit 15	Spare				

SCADA Status Word (Adroit Marshal Agent)		Alarmed	Advanced	Standard	Basic
Bit 0	Spare				
Bit 1	Physical Field Output Value		x	x	x
Bit 2	Output Status Process Value		x	x	x
Bit 3	Spare				
Bit 4	Spare				
Bit 5	Spare				
Bit 6	Fault Active	x	x	x	x
Bit 7	Simulation Enabled		x	x	x
Bit 8	Output Inverted		x	x	x
Bit 9	Delayed On		x	x	
Bit 10	Delayed Off		x		
Bit 11	Spare				
Bit 12	Spare				
Bit 13	Spare				
Bit 14	Alarm bit 0	x	x	x	x
Bit 15	Alarm bit 1	x	x	x	x

Resources	Advanced	Standard	Basic
Digital Inputs	0	0	0
Digital Outputs	1	1	1
Analog Inputs	0	0	0
Analog Outputs	0	0	0
PLC Memory Steps	141	121	105
System Words Used	2	1	0
System Bits Used	64	61	59
SCADA SCAN Points	4	3	2

Template Graphic: Advanced Alarms & Events Screen	Control	Descriptions
	Alarms	Display all the alarms filtered for the current Digital Output
	Events	Display all the events filtered for the current Digital Output.
	Local Acknowledge	Acknowledge all the alarms on locally on the current machine
	Global Acknowledge	Acknowledge all the alarms on globally on all the machines.

Template Graphic: Advanced Trend Screen	Control	Descriptions
	Equipment Name	Display the name of the current digital Output of the trend
	Raw	Display the raw digital output value
	Process Value	Display the processed digital output value